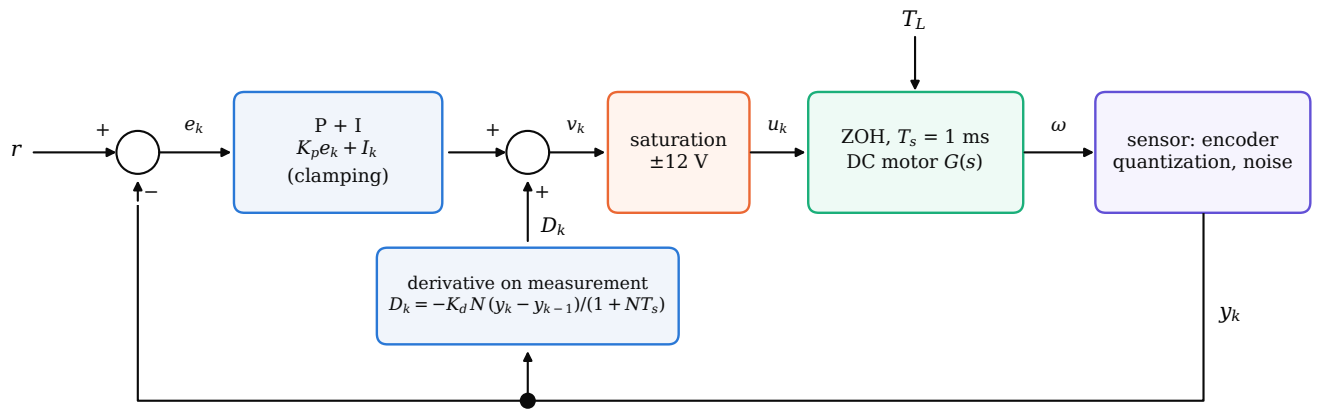
T_L
r
+
−
e_k
P + I
$K_p e_k + I_k$
(clamping)
+
+
v_k
saturation
±12 V
u_k
ZOH, T_s = 1 ms
DC motor $G(s)$
ω
sensor: encoder
quantization, noise
y_k
D_k
derivative on measurement
$D_k = -K_d N (y_k - y_{k-1})/(1 + N T_s)$